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Large deviations methods and the join-the-shortest-queue model

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Abstract

We develop a methodology for studying "large deviations type" questions. Our approach does not require that the large deviations principle holds, and is thus applicable to a larg class of systems. We study a system of queues with exponential servers, which share an arrival stream. Arrivals are routed to the (weighted) shortest queue. It is not known whether the large deviations principle holds for this system. Using the tools developed here we derive large deviations type estimates for the most likely behavior, the most likely path to overflow and the probability of overflow. The analysis applies to any finite number of queues. We show via a counterexample that this sytem may exhibit unexpected behavior.

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